Simulation and Modeling

Course Title: Simulation and Modeling
Course No: BIT407
Nature of the Course: Theory + Lab
Semester: VII

Full Marks: 60 + 20 + 20
Pass Marks: 24 + 8 + 8
Credit Hrs: 3

Course Description:
This course focuses on introducing modeling and simulation of different types of systems, its validation, verification and analysis of simulation output. It also emphasizes on random number and queuing theory.

Course Objectives:
To make students understand the concept of simulation and modeling of real time systems.

Course Contents:

Unit 1: Introduction to Simulation (7 Hrs.)
System Concept, System Modeling, Mathematical Models: Nature and Assumptions, Verification, Validation, and Calibration of Model, Model Development Life Cycle, Advantages and Disadvantages of Simulation, Limitation of the Simulation Techniques, Areas of Applications, Monte Carlo Simulation method

Unit 2: Continuous Systems (8 Hrs.)
Continuous System Model, Differential Equation, Analog Method, Hybrid Computers, Digital-Analog Simulators, Continuous System Simulation Languages (CSSLs), CSMP III, Hybrid Simulation, Feedback Systems, Examples

Unit 3: Discrete System Simulation (9 Hrs.)
Discrete Events, Representation of Time, Generation of Arrival Patterns, Simulation of a Telephone System, Delay Calls, Simulation of Programming Tasks, Gathering Statistics, Counters and Summary Statistics, Measuring Utilization and Occupancy, Recording Distribution and Transit Times, Discrete Simulation Languages

Unit 4: Queuing System (6 Hrs.)
Characteristics and Structure of Basic Queuing System, Models of Queuing System, Queuing Notation, Single Server and Multiple Server Queuing Systems, Measurement of Queuing System Performance, Elementary Idea about Networks of Queuing with Particular Emphasis to Computer System, Applications of Queuing System

Unit 5: Verification and Validation (4 Hrs.)
Design of Simulation Models, Verification of Simulation Models, Calibration and Validation of the models, Three-Step Approach for Validation of Simulation Models, Accreditation of Models

Unit 6: Analysis of Simulation Output (4 Hrs.)
Confidence Intervals and Hypothesis Testing, Estimation Methods, Simulation Run Statistics, Replication of Runs, Elimination of Initial Bias

Unit 7: Simulation Languages (7 Hrs.)
Types of Simulation Languages, Discrete Systems Modeling and Simulation With GPSS, Resources in GPSS, GPSS Programs, Applications, Structural Data and Control Statements, Hybrid Simulation, Feedback Systems : Typical Applications, SIMSCRIPT Programs

Laboratory Works:
Practical should include the simulation of some real time systems (continuous and discrete event systems), Queuing Systems, study of Simulation Tools and Language

References:

1. Jerry Banks, John S. Carson, Barry L. Nelson, David M. Nicole, "Discrete Event system simulation", 5th Edition, Pearson Education
2. Geoffrey Gordon: System Simulation, Prentice Hall of India
3. A.M Law and W.D. Kelton, Simulation Modeling and Analysis, McGraw Hill.
4. A.M Law and R.F. Perry, Simulation: A Problem-solving approach, Addison Wesley Publishing Company.